LAHORE COLLEGE FOR WOMEN UNIVERSITY

DEPARTMENT OF SOFTWARE ENGINEERING

PROJECT

Course: Software Requirements Engineering

Name: Maryam Ghaffar

Roll Number: 24251652035

Section: C

Session: 2024-2028

Submitted To: Mam Ayesha Rafiq

Software Requirements Specification for Online Banking System

Table of Contents

1. Introduction

1.1 Purpose
1.2 Document Conventions
1.3 Intended Audience and Reading Suggestions
1.4 Project Scope
1.5 References

2. Overall Description

2.1 Product Perspective
2.2 Product Features
2.3 User Classes and Characteristics
2.4 Operating Environment
2.5 Design and Implementation Constraints
2.6 User Documentation
2.7 Assumptions and Dependencies

3. System Features

3.1 User Authentication and Login
3.1.1 Description and Priority
3.1.2 Stimulus/Response Sequences
3.1.3 Functional Requirements
3.2 Account Summary and Balance Inquiry
3.2.1 Description and Priority
3.2.2 Stimulus/Response Sequences
3.2.3 Functional Requirements
3.3 Funds Transfer
3.3.1 Description and Priority
3.3.2 Stimulus/Response Sequences
3.3.3 Functional Requirements
3.4 Bill Payment
3.4.1 Description and Priority
3.4.2 Stimulus/Response Sequences
3.4.3 Functional Requirements
3.5 Transaction History
3.5.1 Description and Priority
3.5.2 Stimulus/Response Sequences
3.5.3 Functional Requirements
3.6 Profile Management
3.6.1 Description and Priority

3.6.2 Stimulus/Response Sequences
3.6.3 Functional Requirements
3.7 Customer Service and Support
3.7.1 Description and Priority
3.7.2 Stimulus/Response Sequences
3.7.3 Functional Requirements
3.8 Security and Fraud Monitoring
3.8.1 Description and Priority
3.8.2 Stimulus/Response Sequences
3.8.3 Functional Requirements
4. External Interface Requirements
4.1 User Interfaces
4.2 Hardware Interfaces
4.3 Software Interfaces
4.4 Communications Interfaces
5. Other Nonfunctional Requirements
5.1 Performance Requirements
5.2 Safety Requirements
5.3 Security Requirements
5.4 Software Quality Attributes
6. Other Requirements
Appendix A: Glossary
Appendix B: Analysis Models
Appendix C: Issues List

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to describe the requirements for the **Online Banking System**. This system enables bank customers to perform financial transactions and manage their accounts through a secure web and mobile interface. The SRS covers the full scope of the online banking application for the initial release (version 1.0) and specifies functional and non-functional requirements in detail. This document provides a clear and comprehensive description of the software to be developed, to guide stakeholders, developers, and testers throughout the project lifecycle.

1.2 Document Conventions

This SRS follows standard IEEE formatting conventions as exemplified in this template. Section titles are numbered for clarity according to the template's structure. Requirement statements use the word "shall" to denote mandatory functionality and each requirement is tagged with a unique identifier (e.g., **REQ-1**). Bold text is used for requirement identifiers and emphasis, and code or UI element names are shown in monospace font. Bulleted or numbered lists are used to enumerate features and steps where appropriate.

1.3 Intended Audience and Reading Suggestions

This document is intended for all stakeholders involved in the Online Banking System:
Bank Management and Product Owners: For understanding system capabilities, scope, and compliance considerations.
Software Developers and Architects: For detailed guidance on system requirements, to design and implement the solution.
Quality Assurance and Testers: For planning verification and validation activities based on specified requirements.

End Users (Customers): To understand how they can use the system's features and what functionality is available.
Regulatory and Compliance Personnel: To verify how the system meets legal, security, and privacy obligations.

Readers may proceed sequentially through the sections for a complete understanding. Developers and technical staff should focus on the **System Features** and **External Interface Requirements**, while business stakeholders may concentrate on the **Purpose**, **Project Scope**, and summary of **Product Features**. The **Glossary** in Appendix A provides definitions of specialized terms used in this document.

1.4 Project Scope

The Online Banking System is a secure, web-based and mobile platform provided by the bank to allow customers to manage their financial accounts remotely. It supports account balance inquiries, funds transfers, bill payments, transaction history review, and profile management. By offering these services online, the system aims to improve customer convenience and reduce in-person branch traffic. The system is designed to integrate seamlessly with the bank's existing core banking system, providing a user-friendly interface and robust security. Key objectives include ensuring 24/7 system availability, real-time transaction processing, and compliance with financial regulations to provide reliable and accessible banking services.

1.5 References

- **IEEE Std 830-1998**, "IEEE Recommended Practice for Software Requirements Specifications".
- **PCI DSS (Payment Card Industry Data Security Standard)** v3.2.1, PCI Security Standards Council.
- **ISO/IEC 27001:2013**, "Information Security Management – Requirements".

- **GDPR (General Data Protection Regulation)** (EU Regulation 2016/679).
- Bank internal documents, including the vision and scope statement for the Online Banking project.

2. Overall Description

2.1 Product Perspective

The Online Banking System is a new, self-contained product that provides a digital interface to the bank's existing core banking system. It does not replace the core system but connects to it via secure application programming interfaces (APIs) and database queries. From the user's perspective, the online system appears as the primary way to access account information and perform transactions over the Internet. Internally, the system is part of the bank's suite of digital services and will coexist with other channels (e.g., physical branches, ATMs).

2.2 Product Features

The Online Banking System provides the following high-level features:
User Authentication and Login: Secure login, logout, and multi-factor authentication for account access.
Account Summary and Balances: Display summary information for all user accounts, including current and available balances.
Funds Transfer: Allow users to transfer funds between their accounts and to other accounts within and outside the bank.
Bill Payment: Enable payment of recurring and one-time bills to registered billers (e.g., utilities, credit cards).
Transaction History: Provide detailed history of transactions for each account, with filtering and export options.
Profile Management: Allow users to view and update their personal information and security settings.
Notifications and Alerts: Notify users of important events such as low balances or large transactions (via email or SMS).
Customer Service Support: Provide access to help resources and support channels (e.g., online chat, messaging).
Security and Fraud Monitoring: Continuous monitoring of transactions and activities for suspicious behavior and automatic fraud detection.

These features collectively enable customers to manage their finances online in a comprehensive and secure manner.

2.3 User Classes and Characteristics

The Online Banking System is intended for the following user classes:
Bank Customers (Account Holders): Individuals or businesses who use the system to manage their accounts. They may have varying levels of technical expertise and require a user-friendly interface. They commonly perform transactions like transfers, bill payments, and balance checks.
Potential Customers (Guests): Visitors who may explore the bank's online services or register for online banking. They have not yet been authenticated and have limited access (e.g., to marketing or account sign-up pages).
Bank Employees (Customer Service Representatives, Tellers): Authorized bank staff who use a separate interface to assist customers and manage accounts. They have technical training and require additional privileges (for customer lookups, transaction approvals, etc.).
System Administrators: Technical personnel responsible for maintaining the Online Banking System infrastructure. They perform system setup, monitoring, maintenance, and data backups. They require high-level administrative access to the system configuration and logs.
Regulators and Auditors: Compliance officers or external auditors who may review the system's logs,

reports, and processes to ensure adherence to laws and regulations. They require read-only access to audit trails, security reports, and system documentation.

Each user class has different needs and permissions. For example, customers require intuitive navigation and help support, whereas administrators need advanced system controls and monitoring tools.

2.4 Operating Environment

The Online Banking System will operate in the following environment:
Server-Side: The application will run on the bank's secure server infrastructure or private cloud. The backend includes web/application servers and a relational database (e.g., Oracle or PostgreSQL) to store account data and transaction records.
Client-Side: Customers will access the system through standard desktop or laptop computers, tablets, or smartphones. Supported operating systems include Windows, macOS, Linux (desktop) and iOS/Android (mobile).
Web Browsers: On desktop devices, the system supports modern browsers (latest versions of Chrome, Firefox, Safari, Edge). The web interface is responsive and adapts to various screen sizes. For mobile, either a mobile-optimized web UI or native mobile apps will be provided.
Network: The system communicates over the Internet (TCP/IP) using secure protocols (HTTPS/TLS). Internal connections (e.g., to the core banking system) may use a secure bank intranet or VPN.
Peripheral/Software Components: Users need Internet connectivity and may use hardware tokens or authenticator apps for two-factor authentication. The server may interface with external services such as email servers (SMTP), SMS gateways, and payment network interfaces.

2.5 Design and Implementation Constraints

The following constraints apply to the design and implementation of the Online Banking System:
Regulatory and Security Standards: The system must comply with relevant banking regulations and security standards, including PCI DSS (for payment card data), ISO/IEC 27001 (for information security), and local laws (KYC/AML regulations, GDPR for data privacy).
Technology Stack: The bank may mandate specific technologies (e.g., Java or .NET for server code, React/Angular for UI). All components must adhere to the bank's technical standards, including approved development frameworks and databases.
Authentication Integration: The system must integrate with the bank's existing identity management or single-sign-on (SSO) system (e.g., LDAP or OAuth). Multi-factor authentication (MFA) via SMS or authenticator app must be supported.
Performance and Scalability: The application must support high loads of concurrent users. Scalable architecture (such as clustered servers and load balancing) should be employed to meet performance targets.
Data Storage: Bank data is sensitive. The database design must use encrypted storage as required, and backups must be performed regularly per bank policy.
Concurrency and Atomicity: Transaction processes must be atomic and consistent. The system must integrate with the core banking transaction engine to ensure data integrity (e.g., using ACID transactions).
User Interface Standards: If the bank has a corporate UI style guide or brand guidelines, the online banking UI must comply (logo, colors, typography).
External Dependencies: The system depends on the availability of external systems (e.g., core banking APIs, payment gateways, biller systems). Any limitations or protocols of these systems (such as data formats or throughput limits) will constrain the design.

2.6 User Documentation

The following user documentation will be provided:
User Guide/Manual: A comprehensive guide for customers explaining how to use each feature (with screenshots, step-by-step instructions, and troubleshooting tips). Available as a downloadable PDF and as integrated online help.

Quick-Start Guide: A concise guide for first-time users to quickly set up their account and perform basic tasks (login, view balance, make a transfer).
Online Help & FAQs: Context-sensitive help integrated into the system, including a searchable FAQ section, tooltips, and a knowledge base covering common issues (password reset, transaction FAQs, etc.).
Administrator Guide: Documentation for bank staff and system administrators detailing setup, configuration, maintenance procedures, and administrative functions.
Release Notes: Documentation delivered with each software release, summarizing new features, enhancements, and any bug fixes.

2.7 Assumptions and Dependencies

The following assumptions and dependencies have been identified:
User Access: It is assumed that end users will have reliable Internet access and compatible devices (PC or smartphone) to use the service.
Account Verification: Customers have completed bank onboarding (KYC) and have valid account credentials before using the Online Banking System.
Legacy System Availability: The core banking system and payment network services (e.g., interbank transfer networks) are assumed to be operational and will provide the necessary APIs for integration.
Third-Party Services: Dependencies include third-party services such as SMS gateway providers and external biller systems. These services are assumed to meet uptime and performance SLAs.
Browser Compatibility: It is assumed that users will use supported, up-to-date browser versions. The system may not function correctly on outdated browsers.
Regulatory Environment: The current financial and data privacy regulations are assumed to remain stable during development. Changes in regulations (e.g., new security requirements) could impact the requirements.
Language and Locale: The primary language is English; it is assumed users are fluent in English. Future support for additional languages may be planned but is not part of the initial release scope.

3. System Features

3.1 User Authentication and Login

3.1.1 Description and Priority

This feature provides secure authentication for users accessing the Online Banking System. It includes functionality for user login, logout, session management, and optional multi-factor authentication. Ensuring that only authorized users can access accounts is critical to security, so this feature is of **High Priority**.

3.1.2 Stimulus/Response Sequences

- **Stimulus:** The user navigates to the login page and enters their username and password.
  Response: The system verifies the credentials against the user database. If the credentials are valid, the system proceeds with login; if invalid, it displays an error message and prompts the user to re-enter credentials.
- **Stimulus:** After successful credential verification, if multi-factor authentication (MFA) is enabled for the user, the system sends a one-time code (via SMS or email).
  Response: The user enters the one-time code on the prompted page. The system validates the code. If correct, the user is granted access and a session is established; if incorrect, the user is prompted to retry or request a new code.
- **Stimulus:** The user chooses to log out or closes the browser.
  Response: The system terminates the user session and redirects to the login page, ensuring no further access without re-authentication.
- **Stimulus:** The user remains idle on an authenticated page for a predefined timeout period (e.g., 15 minutes).

Response: The system automatically ends the session and requires re-authentication, displaying a timeout notice to the user.

3.1.3 Functional Requirements

- **REQ-1:** The system shall require every user to authenticate with a unique username and password before accessing any other functionality.
- **REQ-2:** The system shall allow users to reset their password through a secure recovery process (e.g., sending a reset link to the user's registered email).
- **REQ-3:** The system shall support multi-factor authentication (MFA) by sending a one-time code (OTP) to the user's registered mobile number or email after the password is verified.
- **REQ-4:** The system shall lock a user's account for a configurable period after a specified number of consecutive failed login attempts, and notify the user of the lockout via email or SMS.
- **REQ-5:** The system shall automatically log out users after a period of inactivity (e.g., 15 minutes) and require re-authentication for any new actions.
- **REQ-6:** The system shall use secure protocols (e.g., HTTPS/TLS) for all authentication-related data transmission and shall never transmit passwords in plain text.

3.2 Account Summary and Balance Inquiry

3.2.1 Description and Priority

This feature allows authenticated users to view a summary of their accounts, including current and available balances. It provides a dashboard or account overview page that lists each account (checking, savings, loans, etc.) and shows key information such as account type, partial account number, and balances. Users can quickly assess their financial status. This feature is of **High Priority** because it is a core function of online banking.

3.2.2 Stimulus/Response Sequences

- **Stimulus:** The user selects the "Account Summary" or "Dashboard" option after logging in.
  Response: The system retrieves the user's account information from the core banking system and displays a summary list of all accounts with current balance and available balance for each.
- **Stimulus:** The user clicks on a specific account in the summary list.
  Response: The system displays detailed information for that account, including recent transactions and additional account details.
- **Stimulus:** The user refreshes the account summary page.
  Response: The system updates the displayed balances to reflect any new transactions that have completed since the page was last loaded.

3.2.3 Functional Requirements

- **REQ-7:** The system shall display a list of all bank accounts associated with the authenticated user.
- **REQ-8:** For each account listed, the system shall show the account type (e.g., checking, savings), account number (partially masked for security), current balance, and available balance.
- **REQ-9:** The system shall allow the user to select an account to view more detailed information, such as a recent transaction list or account details.
- **REQ-10:** The system shall indicate any special status on an account (e.g., overdraft protection, holds, or account restrictions) if applicable.
- **REQ-11:** The system shall update account balances in real time or near-real time after any completed transactions, so that users always see current information.

3.3 Funds Transfer

3.3.1 Description and Priority

This feature enables users to transfer funds between accounts. It supports transfers between the user's own accounts (internal transfers) as well as transfers to other accounts (either at the same bank or at other banks). Scheduled and recurring transfers may also be included. The feature includes validating available funds and compliance with transfer limits. Due to its importance for moving money, this feature is of **High Priority**.

3.3.2 Stimulus/Response Sequences

- **Stimulus:** The user navigates to the "Transfer Funds" section, selects a source account, a destination account (or enters a new payee), and enters the transfer amount.
  Response: The system validates the input (checking that the destination account exists and the source account has sufficient funds) and displays a summary of the transfer details, asking for user confirmation.
- **Stimulus:** The user confirms the transfer request.
  Response: The system processes the transfer by communicating with the core banking system (and possibly external payment networks for interbank transfers). The source account is debited and the destination account is credited. A success or failure notification (with a transaction reference number) is displayed to the user.
- **Stimulus:** The user attempts to transfer more funds than are available in the source account.
  Response: The system detects insufficient funds, does not execute the transfer, and displays an error message informing the user.
- **Stimulus:** The user chooses to schedule a future or recurring transfer, specifying details (amount, frequency, start date).
  Response: The system captures the scheduling details and sets up the transfer to execute at the specified times, providing confirmation to the user that the recurring transfer is scheduled.

3.3.3 Functional Requirements

- **REQ-12:** The system shall allow users to transfer funds from any of their accounts to any other account within the same bank (including accounts of other users if allowed).
- **REQ-13:** The system shall allow users to transfer funds to accounts at other banks (interbank transfers) using standard payment methods (e.g., ACH, wire transfer), subject to bank policy and transfer limits.
- **REQ-14:** The system shall require the user to confirm the transfer details (source account, destination account, amount) before executing the transfer.
- **REQ-15:** The system shall check that the source account has sufficient funds (including any required minimum balances) before proceeding; if not sufficient, the system shall abort the transfer and notify the user.
- **REQ-16:** The system shall record each successful transfer with a unique transaction reference number, timestamp, and relevant details (source, destination, amount) in the transaction history.
- **REQ-17:** The system shall allow users to schedule one-time future transfers or recurring transfers (e.g., weekly or monthly) by specifying the date(s) and frequency.
- **REQ-18:** The system shall enforce any per-transaction or daily transfer limits as defined by bank policy, and shall prevent transfers that exceed these limits, notifying the user accordingly.

3.4 Bill Payment

3.4.1 Description and Priority

This feature allows users to pay bills from their accounts to registered billers (such as utilities, credit card companies, and other service providers). Users can add new billers, view bill details, and schedule payments.

Bill payments may be one-time or recurring. This feature is of **High Priority** because it provides essential convenience and encourages continued use of the online banking platform.

3.4.2 Stimulus/Response Sequences

- **Stimulus:** The user selects the "Bill Payment" section and chooses an existing payee or adds a new payee by entering billing details (e.g., biller name, billing account number).
  Response: For a new payee, the system verifies the entered details (where possible) and, if valid, adds the payee to the user's saved payee list. For an existing payee, the system displays the current payment details.
- **Stimulus:** The user enters the payment amount and selects a payment date (immediate or future).
  Response: The system verifies that the user's account has sufficient funds, then displays a payment summary (including payee name and payment amount) for confirmation.
- **Stimulus:** The user confirms the payment.
  Response: The system processes the payment by debiting the user's account and sending the payment instruction to the biller or payment clearing system. The system then displays a confirmation with a payment ID to the user.
- **Stimulus:** The user schedules a future bill payment.
  Response: The system saves the scheduled payment details. On the specified date, the system executes the payment and sends a notification to the user. The user can view upcoming scheduled payments in their profile.

3.4.3 Functional Requirements

- **REQ-19:** The system shall allow users to add new biller payees by entering necessary information (such as biller name, account number). The system shall validate required fields.
- **REQ-20:** The system shall enable users to initiate one-time or recurring bill payments to any saved payee.
- **REQ-21:** The system shall display a summary of the payment details (biller name, amount, date) and require the user to confirm before executing the payment.
- **REQ-22:** The system shall generate and display a confirmation for each successful bill payment, including a payment reference number, date, and biller name. It shall also send this confirmation via the user's preferred notification method (email or SMS).
- **REQ-23:** The system shall prevent bill payment if the user's account balance is insufficient, and shall display an error message indicating the failed payment.

3.5 Transaction History

3.5.1 Description and Priority

This feature provides users with access to a historical record of transactions for each of their accounts. Users can view past deposits, withdrawals, payments, and transfers, filter results, and export statements. This is a **High Priority** feature because it is essential for users to review their financial activity and track spending.

3.5.2 Stimulus/Response Sequences

- **Stimulus:** The user navigates to the account details page and selects the "View History" or "Statement" option.
  Response: The system retrieves and displays a list of past transactions for that account, typically covering recent history (e.g., the last 3-12 months) or as specified by the user. Each entry shows date, description, amount, and balance.
- **Stimulus:** The user applies a date range filter or transaction type filter (e.g., show only debits).
  Response: The system refreshes the transaction list to show only entries matching the selected criteria (within the date range, of the specified type).

- **Stimulus:** The user opts to download or print an account statement.
  Response: The system generates a statement (e.g., as a PDF or CSV file) containing transactions for the chosen date range and prompts the user to save or print the document.

3.5.3 Functional Requirements

- **REQ-24:** The system shall provide a chronological list of all transactions for a selected account, including date, description, debit/credit amount, and resulting balance.
- **REQ-25:** The system shall allow users to filter the transaction history by date range and/or transaction type (such as debit, credit, or specific transaction categories).
- **REQ-26:** The system shall allow users to search the transaction history by keyword (e.g., merchant name) or amount.
- **REQ-27:** The system shall enable users to download or email account statements for a specified date range in a common format (e.g., PDF, CSV).
- **REQ-28:** The system shall clearly indicate any pending or uncleared transactions separate from posted transactions (if applicable).

3.6 Profile Management

3.6.1 Description and Priority

This feature allows users to view and update their personal profile information and security settings. Functions include changing passwords, updating contact information (email, phone number), and configuring notification preferences. This feature is of **Medium Priority** because it enhances user experience and security but is not as critical as transactional features.

3.6.2 Stimulus/Response Sequences

- **Stimulus:** The user selects the “Profile” or “Settings” section in their account menu.
  Response: The system displays the user’s current profile information (name, address, contact details) and available actions (e.g., edit fields, change password, configure alerts).
- **Stimulus:** The user edits a personal detail (such as phone number) and submits the changes.
  Response: The system validates the input (e.g., correct phone number format) and, if valid, saves the change to the user’s profile. It then displays a confirmation message.
- **Stimulus:** The user chooses to change their password.
  Response: The system prompts the user to enter the current password and a new password. The system verifies the current password, checks that the new password meets strength requirements, updates it, and confirms the change to the user.
- **Stimulus:** The user updates notification preferences (e.g., enabling email alerts for transactions).
  Response: The system saves the updated preferences and will use them for future notifications.

3.6.3 Functional Requirements

- **REQ-29:** The system shall allow the user to view and update personal information such as mailing address, phone number, and email address.
- **REQ-30:** The system shall allow the user to change their password; the user must provide the current password, and the new password must meet complexity requirements (e.g., minimum length, character variety).
- **REQ-31:** The system shall allow the user to set or update security preferences, such as enabling/disabling multi-factor authentication and selecting notification methods (email or SMS alerts).
- **REQ-32:** The system shall require additional verification (such as re-entering the current password or entering an OTP) when the user changes critical information (like email address or security settings).

3.7 Customer Service and Support

3.7.1 Description and Priority

This feature provides users with access to customer support resources, including online help, FAQs, and live support channels. It also includes mechanisms for submitting support requests or feedback (e.g., tickets or contact forms). This feature is of **Medium Priority**, as it supports user needs but is ancillary to core banking operations.

3.7.2 Stimulus/Response Sequences

- **Stimulus:** The user selects “Help” or “Support” from the navigation menu.
  Response: The system displays help resources such as FAQs, tutorials, and contact information for customer support (phone numbers, chat links, etc.).
- **Stimulus:** The user initiates a live chat with a customer service representative (if available).
  Response: The system establishes a chat session, optionally authenticates the user to the agent, and allows the user to send messages. If no agent is available, the system offers to schedule a callback or send an email.
- **Stimulus:** The user submits a support request or feedback form.
  Response: The system records the issue details and generates a support ticket with a unique ID. It then displays an acknowledgment with the ticket number and an estimated response time.

3.7.3 Functional Requirements

- **REQ-33:** The system shall provide an online help center with searchable FAQs and user guides relevant to various features of the application.
- **REQ-34:** The system shall allow users to initiate a live chat with a customer support agent. If agents are unavailable, the system shall collect the user’s contact information and issue details for follow-up.
- **REQ-35:** The system shall provide a form for users to submit support requests or feedback. Upon submission, the system shall generate a support ticket number and display a confirmation message to the user.
- **REQ-36:** The system shall display context-sensitive help links or tooltips on relevant pages (for example, explanations next to form fields or icons).

3.8 Security and Fraud Monitoring

3.8.1 Description and Priority

This feature involves monitoring user activities and transactions to detect and prevent fraudulent behavior. It includes real-time checks for unusual patterns (e.g., large transfers, rapid login attempts) and the ability to flag or block suspicious activity. Ensuring the security of customer assets is of **High Priority**, so this feature is critical to the system.

3.8.2 Stimulus/Response Sequences

- **Stimulus:** The user performs a suspicious action (e.g., several failed login attempts or a transfer exceeding a normal threshold).
  Response: The system flags the behavior and takes action (such as temporarily locking the account or requiring additional authentication). It may notify the user of the suspicious activity and prompt them to confirm recent actions.
- **Stimulus:** The system detects a security event (e.g., known vulnerability scan or intrusion attempt).
  Response: The system logs the event with high priority and triggers security protocols (such as activating firewalls, alerting administrators, or isolating affected components).

- **Stimulus:** A password change or login from a new device occurs on a user's account.
  Response: The system sends a notification (via email or SMS) to the user's registered contact information to confirm the action, providing details like date/time and device location.
- **Stimulus:** The user triggers a transaction (transfer or payment).
  Response: The system evaluates the transaction against fraud detection rules (for example, comparing against historical usage). If the transaction is highly unusual, the system may request additional confirmation or escalate to a fraud review team.

3.8.3 Functional Requirements

- **REQ-37:** The system shall log all user actions (such as login attempts, transfers, profile changes) with timestamps for auditing and review.
- **REQ-38:** The system shall automatically alert the user via email or SMS when critical account events occur (e.g., password changes, large transfers, or logins from a new device) to allow the user to verify their legitimacy.
- **REQ-39:** The system shall implement fraud detection rules (such as limits on login attempts and transaction velocity checks) and take appropriate action (extra authentication, alerts, or lockouts) when thresholds are exceeded.
- **REQ-40:** The system shall encrypt all sensitive data (passwords, personal data, account numbers) both at rest and in transit, using industry-standard encryption (for example, AES-256 for data storage and TLS 1.3 for communication).
- **REQ-41:** The system shall enforce role-based access control for all internal users (bank staff), ensuring that each user can only access data and functions appropriate to their role.

4. External Interface Requirements

4.1 User Interfaces

- **Web Interface:** The system shall provide a responsive web interface accessible via desktop and mobile browsers. This interface will include screens for login, dashboard (account summary), funds transfer, bill payment, transaction history, profile settings, and help. The UI shall be intuitive and consistent with the bank's visual style guide (including colors, logos, and fonts).
- **Mobile Application (Optional):** If provided, native mobile apps for iOS and Android shall be available. They will offer equivalent functionality to the web interface and may utilize mobile features (e.g., push notifications, biometric login with fingerprint or Face ID).
- **Accessibility:** The user interface shall comply with common accessibility standards (such as WCAG 2.1 Level AA) to ensure usability by individuals with disabilities.
- **Language and Localization:** Initially, the UI will support English. The design shall allow easy addition of other languages and regional formats in the future (e.g., displaying amounts with local currency formats).
- **Administrator Interface:** A separate administrative web portal (requiring separate credentials) shall be provided for bank staff. It will allow staff to manage user accounts, view audit logs, and perform necessary back-office operations. This interface will use forms and reports consistent in style with the customer interface.

4.2 Hardware Interfaces

- **User Devices:** The system will not depend on any specialized client hardware. Users will access the system via standard devices (PCs, laptops, tablets, smartphones) using typical input methods (keyboard, mouse, touch screen).
- **Server Hardware:** On the server side, the system will run on bank-owned servers or cloud instances with sufficient CPU, memory, and storage to meet performance requirements. Standard networking equipment (routers, switches, firewalls) will support connectivity.

- **Security Modules:** The system may interface with hardware security modules (HSMs) for secure encryption key storage and cryptographic operations, as required by the bank's security policies.
- **Transaction Infrastructure:** While not directly controlled by this system, online transactions may interact with external financial networks (e.g., SWIFT or ACH). This involves hardware (switches, routers) in payment networks which are out of scope for direct interface specification.

4.3 Software Interfaces

- **Core Banking System API:** The system shall interface with the bank's core banking software via secure APIs or web services. These interfaces support account lookups, balance inquiries, transaction posting, and other necessary operations. All calls must be authenticated.
- **Payment Gateways:** For interbank transfers and bill payments, the system shall connect to external payment networks or third-party payment gateway services (e.g., ACH/RTGS networks, bill payment processors) using standard protocols.
- **Messaging Services:** The system shall interface with an email server (SMTP) and SMS gateway to send notifications, one-time codes, and alerts to users. It will use secure authentication for these services.
- **Directory/Authentication Service:** If the bank uses an external identity provider (such as LDAP or Active Directory), the system shall integrate with it for user authentication and possibly single sign-on.
- **Logging/Monitoring Tools:** The system shall integrate with enterprise logging and monitoring solutions (e.g., a SIEM system) to report security events and system health metrics.

4.4 Communications Interfaces

- **Network Protocols:** All user access shall occur over the Internet using HTTPS (HTTP over TLS) to ensure encryption. Internal communications (e.g., between application servers and databases) may use secure protocols such as TLS-secured RPC or SSH.
- **Ports and Firewall:** The system shall use standard ports (e.g., 443 for HTTPS). Network firewalls and intrusion prevention systems will be configured to allow only necessary traffic to these ports.
- **Integration Protocols:** The system shall use standard protocols for external integrations, such as REST/JSON or SOAP/XML for web service calls. For batch data exchange (e.g., nightly reports), secure file transfer protocols (e.g., SFTP) may be used.
- **API Security:** All APIs (internal or external) shall require authentication (such as tokens or certificates) and authorization checks. Only authorized systems/users should be able to invoke critical operations.
- **Messaging:** The system shall use secure channels (e.g., SMTPS or STARTTLS for email, TLS for API calls). It may use protocols such as SMPP for sending SMS messages to telecom providers.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall handle at least **1,000 concurrent active users** without performance degradation and scale to **5,000** concurrent users at peak by adding hardware resources.
- Standard user actions (page loads, balance inquiries, initiating transactions) shall have a response time under **2 seconds** under normal conditions.
- The system shall process at least **100 transactions per second** during peak load. End-of-day batch processes (e.g., posting overnight transactions) shall complete within the designated maintenance window (e.g., overnight hours).
- The system shall achieve **99.9% uptime** (excluding scheduled maintenance), corresponding to no more than ~8 hours of downtime per year.

- In case of a server failure, the backup servers or cluster shall failover automatically or within **5 minutes** to meet the availability target.

5.2 Safety Requirements

- The system shall ensure **transactional safety**: each funds transfer or payment must be atomic (either fully complete or not executed) to prevent partial execution.
- In the event of an error (e.g., power loss or network failure during a transaction), the system shall roll back any incomplete transactions to maintain data integrity.
- The system shall maintain regular backups (at least daily) of all critical data. A disaster recovery plan shall exist to restore the system in case of catastrophic failure.
- While the system is not safety-critical in terms of human physical safety, it must maintain **data integrity** and prevent unauthorized or unintended transactions through proper validation and error handling.

5.3 Security Requirements

- **Authentication & Authorization:** The system shall enforce strong authentication (including multi-factor) and role-based access control. User sessions shall be securely managed and expire after inactivity.
- **Data Protection:** All sensitive customer data (e.g., personal information, account details, passwords) shall be encrypted at rest (e.g., database encryption) and in transit (TLS).
- **Privacy & Compliance:** The system shall comply with data protection laws (e.g., GDPR) by safeguarding personal data, allowing users to manage consent, and implementing data retention/deletion policies.
- **Audit and Logging:** All significant events (logins, transactions, password changes, administrative actions) shall be logged in an audit trail. Logs shall be tamper-evident, timestamped, and retained per regulatory requirements.
- **Input Validation:** The system shall validate and sanitize all user inputs to prevent security vulnerabilities (such as SQL injection and cross-site scripting), following OWASP best practices.
- **Network Security:** Communications shall be protected by firewalls and intrusion detection/prevention systems. Regular security assessments (vulnerability scans and penetration tests) shall be conducted.
- **Session Security:** Session identifiers shall be cryptographically secure. The system shall protect against session hijacking (e.g., using secure cookies and not exposing tokens in URLs).
- **Error Handling:** Error messages displayed to users shall not expose sensitive system information (e.g., stack traces or database errors).
- **Security Updates:** The system shall be designed to allow timely application of security patches and updates for all software components to address newly discovered vulnerabilities.

5.4 Software Quality Attributes

- **Reliability:** The system shall operate continuously and correctly during normal use. It shall include automated self-tests and monitoring to detect failures.
- **Maintainability:** The software shall use a modular architecture. Code and system design shall be documented to facilitate future enhancements and maintenance. All code must follow the bank's coding standards.
- **Usability:** The user interface shall be intuitive and consistent. The system shall provide clear navigation and help messages so that users can perform common tasks without extensive training.
- **Scalability:** The system shall accommodate growth in user base and transaction volume by scaling out (adding additional servers or cloud resources) without major redesign.

- **Interoperability:** The system shall operate with standard web browsers and mobile OS versions without requiring special plugins. It shall integrate with external systems using widely accepted protocols.
- **Availability:** (See Performance Requirements) The system shall support high availability. It should quickly recover from failures (e.g., automatic failover for critical components).
- **Portability:** The system's design shall allow deployment on equivalent hardware or cloud environments (e.g., switching to a different cloud provider) with minimal changes.

6. Other Requirements

- **Regulatory Compliance:** The system shall comply with all applicable financial regulations and standards (e.g., KYC/AML rules, PCI DSS). Audit trails and reporting features shall support regulatory audits and investigations.
- **Legal:** The system shall enforce data retention policies as required by law (for example, retaining transaction logs for a minimum number of years). It shall also comply with any directives on user data storage and deletion.
- **Localization:** Initially, the system will support English language only. Support for additional languages and regional settings may be considered in future versions.
- **Service Level Agreement (SLA):** The online banking service shall meet any agreed SLA terms (e.g., 99.9% uptime, response times) as defined by the bank's operations team. Monitoring and reporting mechanisms shall be in place to demonstrate compliance with SLA targets.
- **Business Continuity:** A business continuity plan shall exist to ensure the banking service can continue or be restored rapidly after a disaster (e.g., having off-site backups, redundant data centers, and failover procedures).

Appendix A: Glossary

- **Online Banking System:** A software application provided by the bank that allows customers to perform banking tasks (like transfers, payments, and account monitoring) over the Internet.
- **User Account:** The online banking profile assigned to a customer, which gives them access to their associated bank accounts.
- **Authentication:** The process of verifying the identity of a user, typically through credentials like a username and password, often supplemented by additional factors.
- **Multi-Factor Authentication (MFA):** A security method that requires two or more independent credentials (e.g., password plus one-time code) to verify a user's identity.
- **Account Balance:** The amount of money in a bank account at a given moment (current balance) and the portion of that balance that is available for withdrawal (available balance).
- **Transaction:** An operation that changes the balance of an account, such as a deposit, withdrawal, transfer, or payment.
- **Core Banking System:** The bank's central system where account data is stored and where most banking transactions are processed.
- **PCI DSS:** Payment Card Industry Data Security Standard, a set of requirements for organizations that handle credit card and payment card information.
- **KYC:** Know Your Customer, a process in which a bank verifies the identity and legitimacy of its customers in compliance with regulations.
- **GDPR:** General Data Protection Regulation, a European Union law for data protection and privacy of individuals.

Appendix B: Analysis Models

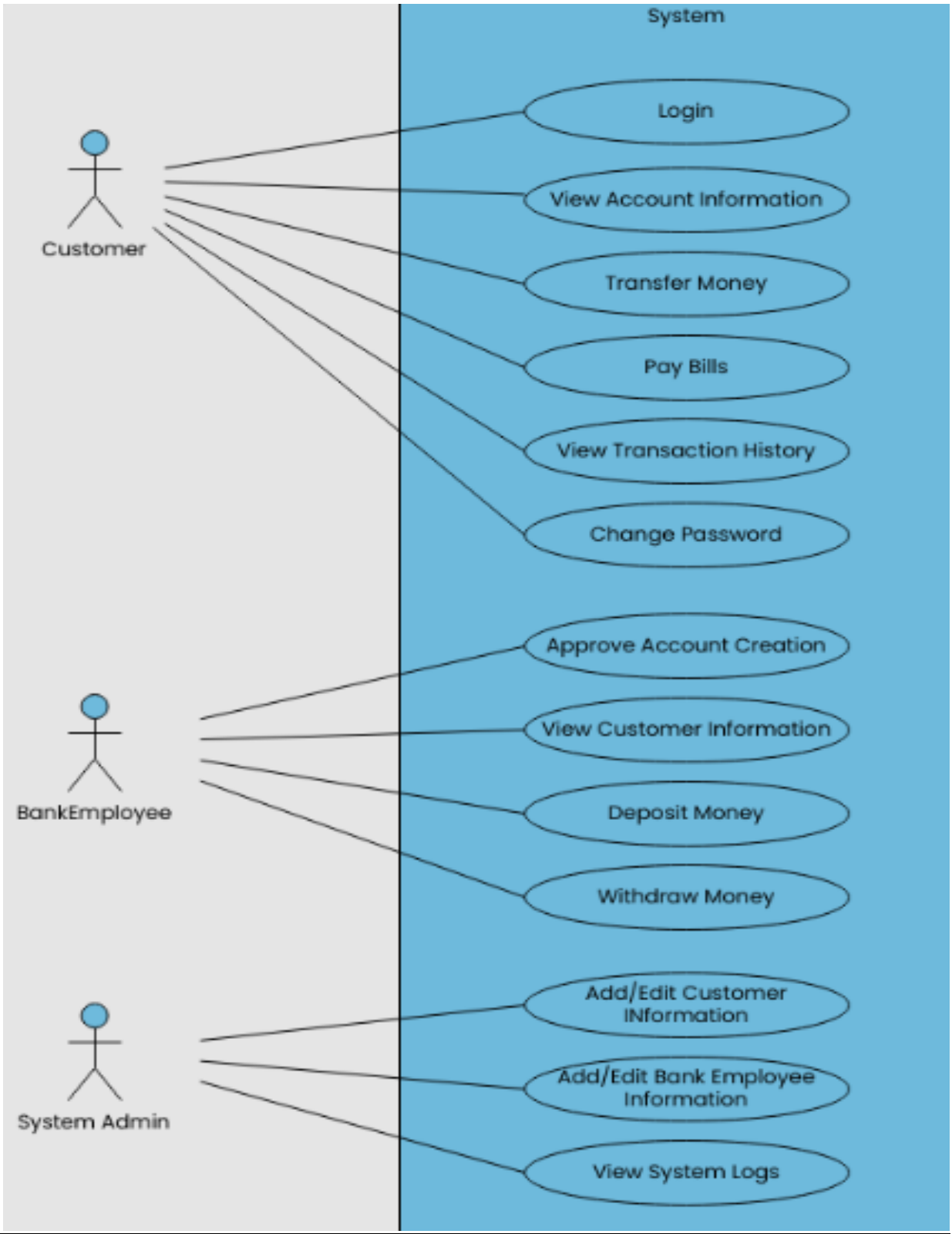
- **B.1 Context Diagram:** Shows the system as a black box and its interactions with external entities (customers, core banking system, payment networks).
- **B.2 Use-Case Diagrams:** Illustrate the main actors (e.g., Customer, Administrator) and their use cases (e.g., Login, Transfer Funds).
- **B.3 Data Flow Diagrams:** Depict how data moves through the system for key processes (e.g., from user input to account update).
- **B.4 Entity-Relationship Diagram:** Shows key data entities (such as Account, User, Transaction) and their relationships.

Appendix C: Issues List

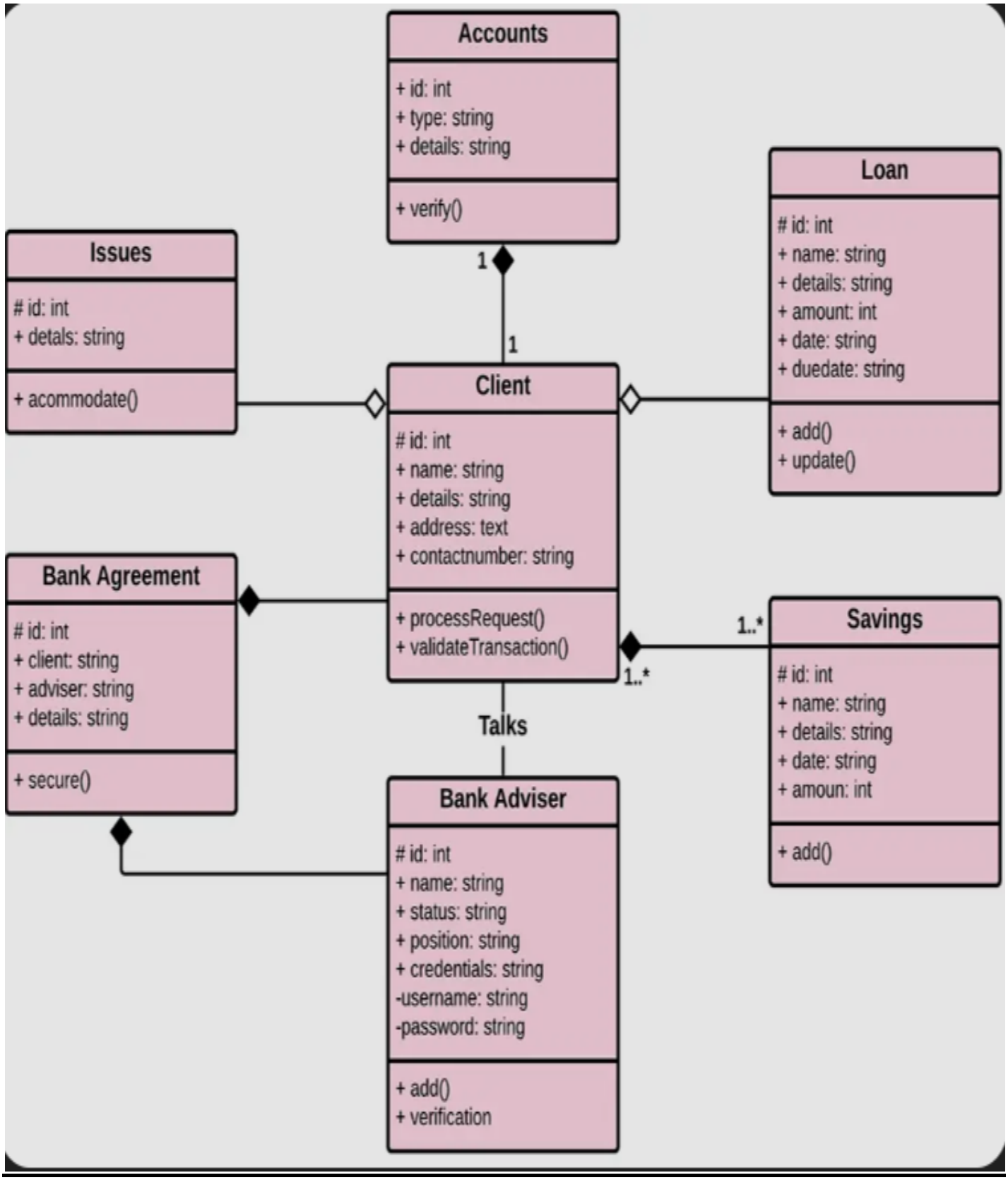
- **I-1:** Determine the level of support for multiple currencies and how exchange rates will be obtained and applied in transfers.
- **I-2:** Confirm the list of billers and the integration method (e.g., APIs) required for bill payment functionality.
- **I-3:** Define the retention period for audit logs and transaction data based on legal and internal policy requirements.
- **I-4:** Finalize SLA parameters with the IT operations team (target availability, peak load handling) and ensure infrastructure meets those targets.
- **I-5:** Decide on implementation details for multi-factor authentication (e.g., SMS vs. authenticator app) and supported mobile platforms.

DIAGRAMS

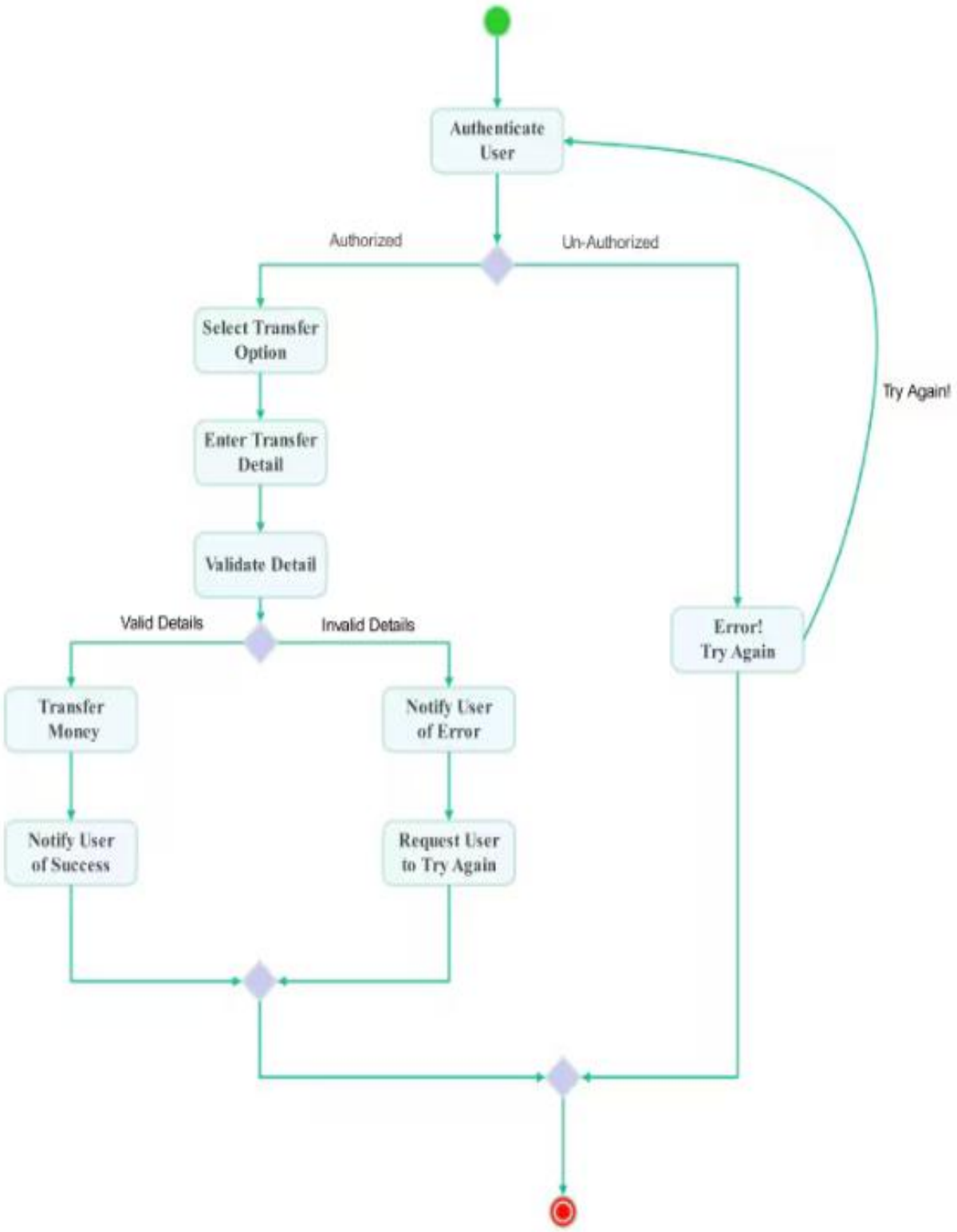
Use Case



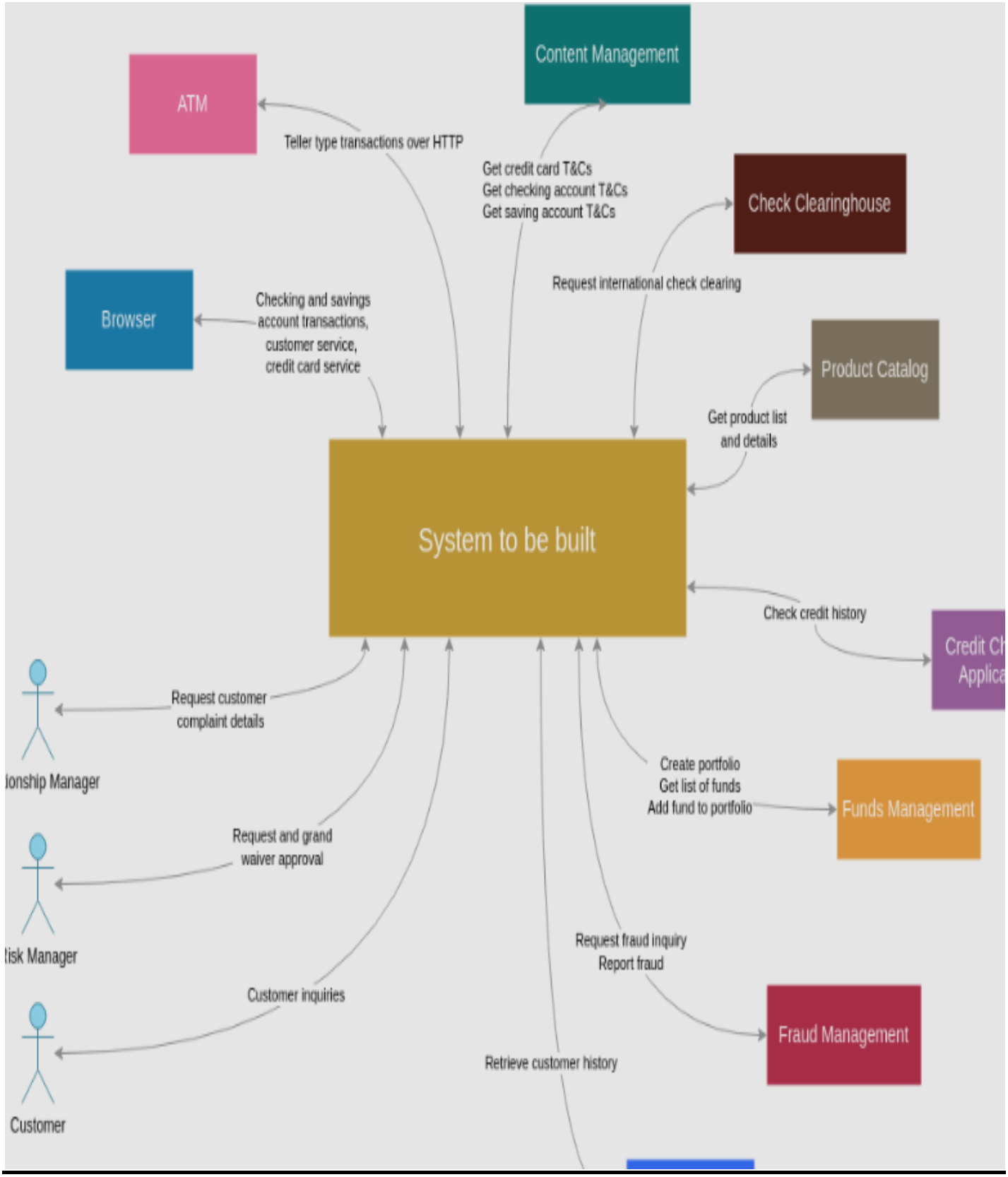
Class Diagram



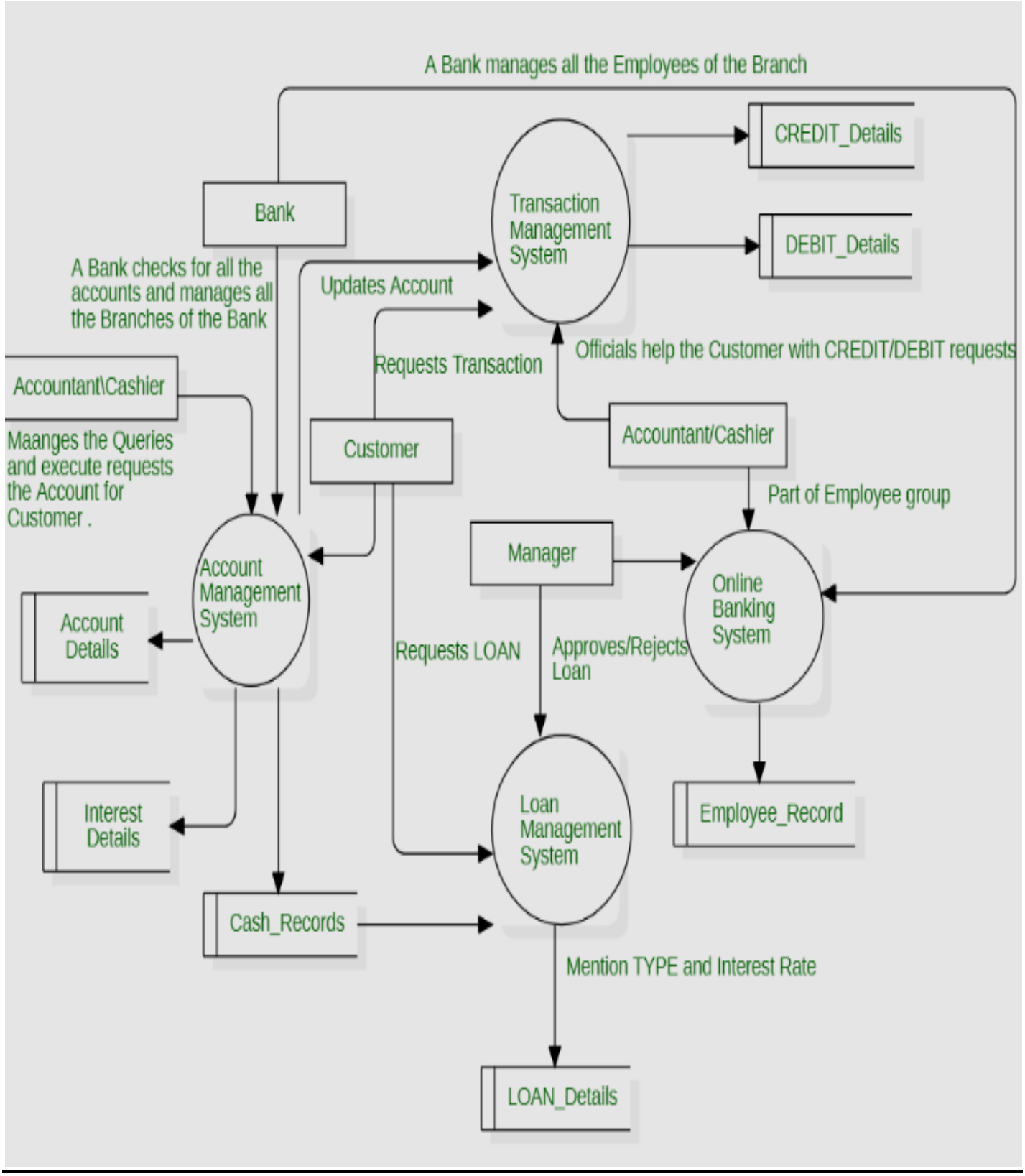
Activity Diagram



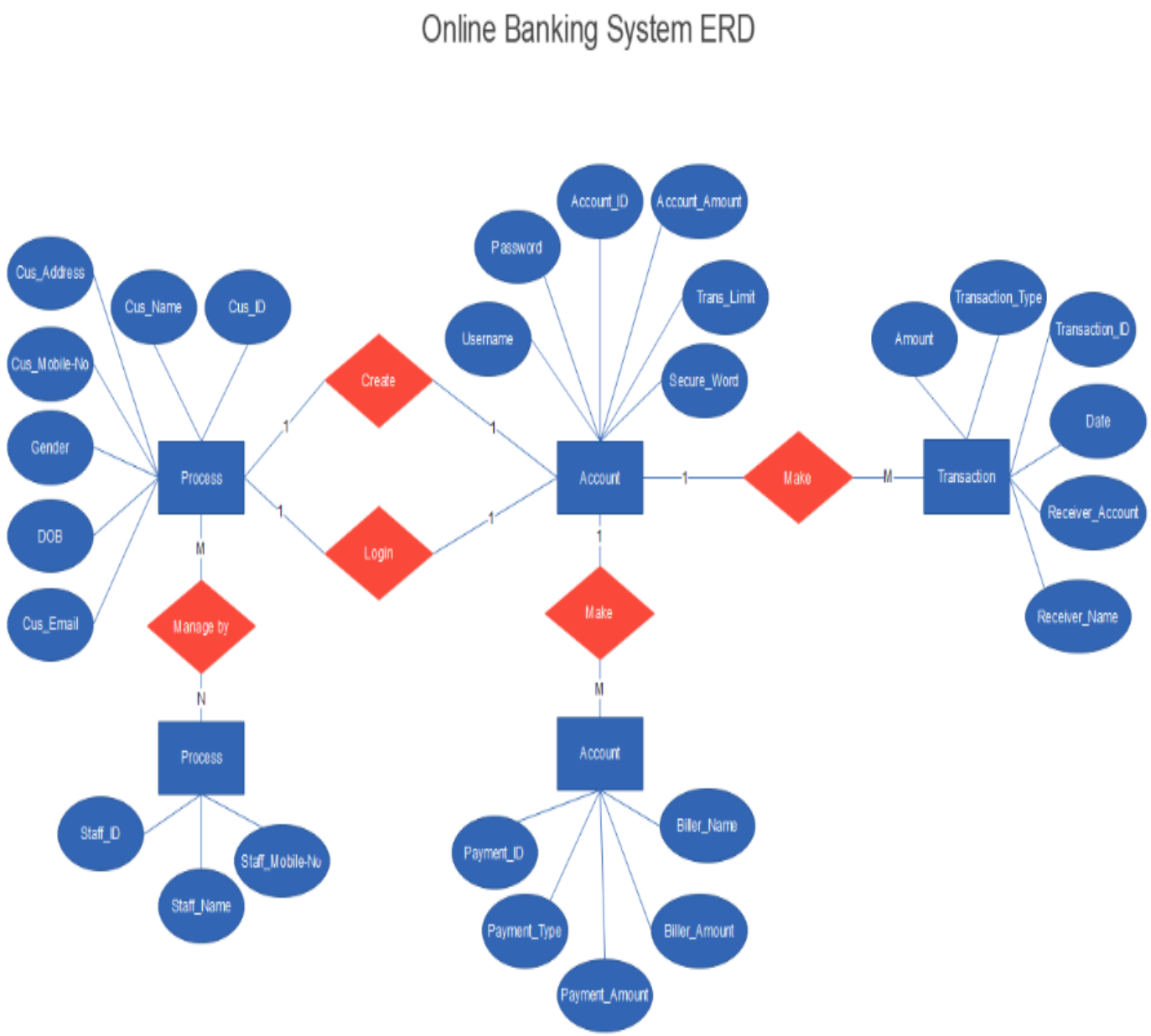
Context Diagram



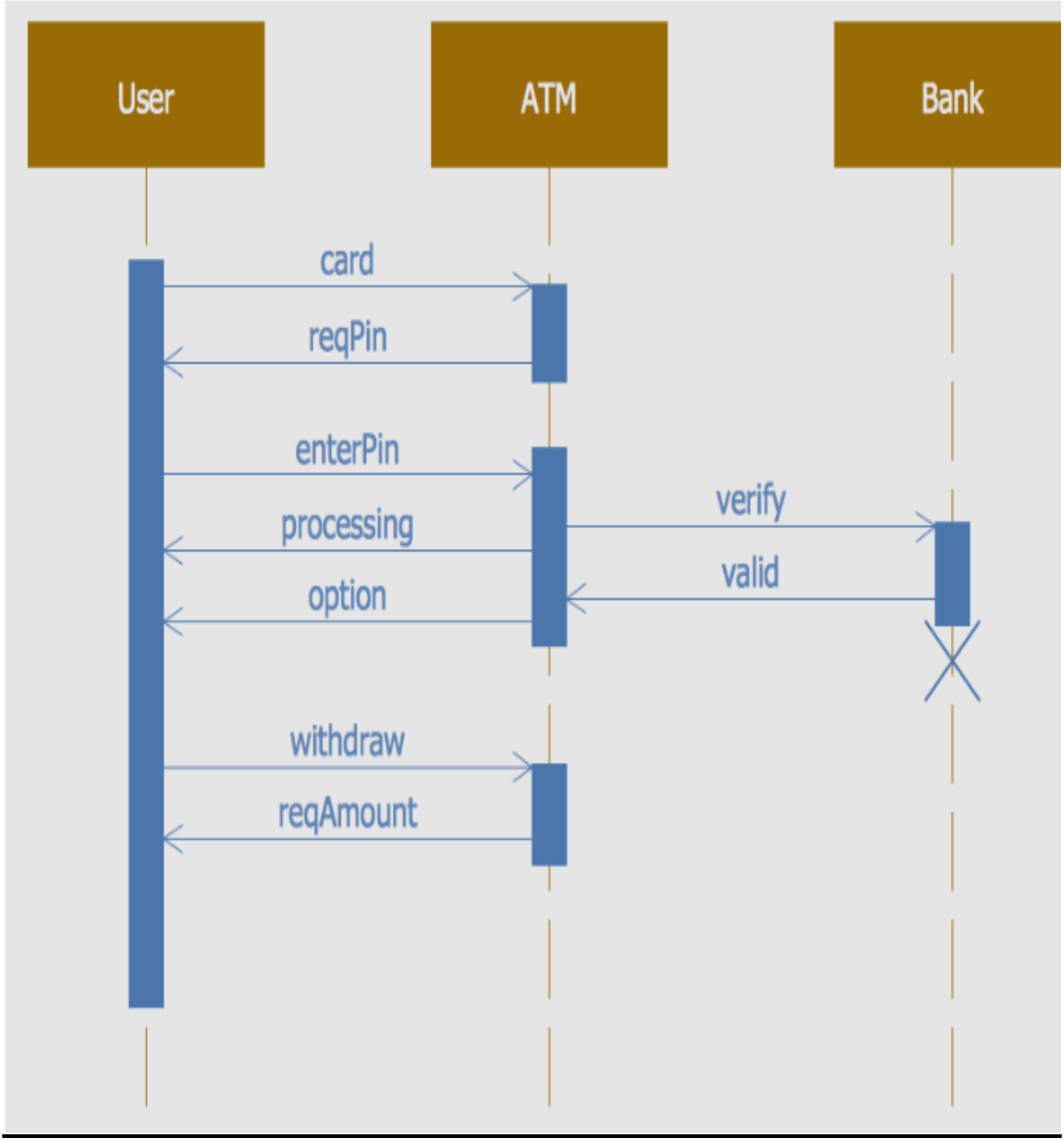
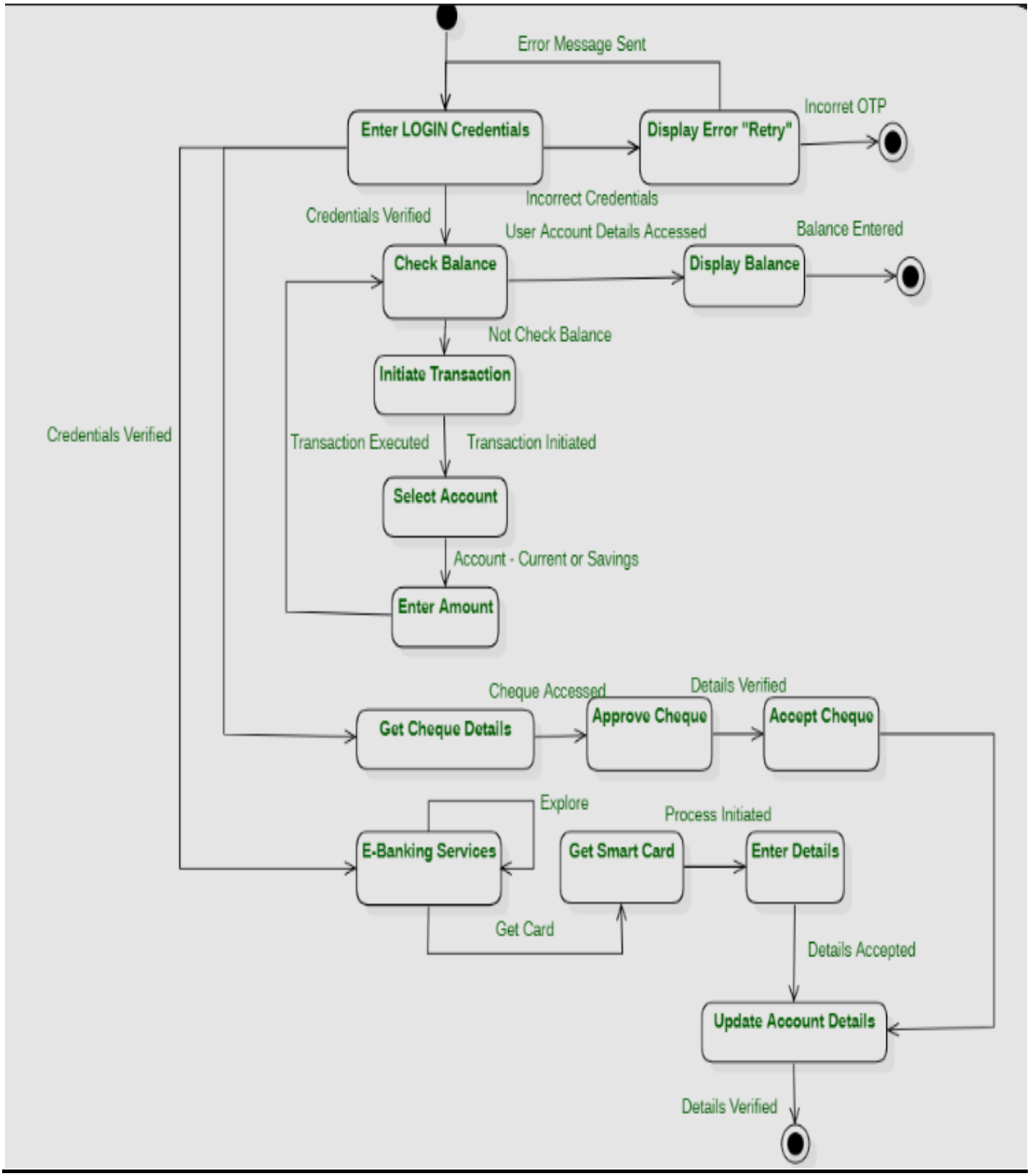
Data Flow Diagram



Entity Relationship Diagram



Sequence Diagram



State Diagram